

4.7 Option 5: Pollution and Human Health at Risk

Pollution and human health

Human health is a key concern at scales ranging from personal to global. Personally health has a key impact on quality of life, but it also affects economic development at broader scales and poses key global challenges relating to the spread of disease. Health risk is strongly related to level of economic development, either in the form of transmissible disease or environmental pollution. The spread of risk follows geographical patterns and features. In modern societies, pollution is a key risk especially in countries where rapid economic development takes precedence over environmental and health concerns.

Risk

A wide range of strategies can be adopted to manage pollution and health; some problems are harder to manage than others and require long-term strategies, economic and lifestyle changes. Increasingly management is international in nature, reflecting an interconnected world.

Global synoptic context

Key linkages

Places

Where are the polluted and unhealthy places?

People

Who suffers as a result of pollution and health risk?

Power

Who is responsible for managing these risks?

Suggested teaching and learning

Mapping areas of air pollution at a global scale, and areas at risk from disease, eg malaria and TB.

Exploring the link between pollution, health and development using a global statistics website such as the World Resources Institute.

Examining the role of a UN organisation such as the WHO.

1 Defining the risks to human health

Enquiry question: *What are the health risks?*

What students need to learn	Suggested teaching and learning
■ Human health risks.	■ Exploring a range of health risks from short term to chronic.
■ Patterns of health risk at different scales (global, national, local).	■ Researching the different geographical patterns of health and disease at different scales.
■ Health risk patterns over time.	■ Investigating how these health risks may alter over time, as shown in the epidemiology model.
■ How health affects both the quality of life and economic development.	■ Developing an awareness of the importance of health to the overall quality of life of people and the economic development of areas/nations and the impacts of poor health.

2 The complex causes of health risk

Enquiry question: *What are the causes of health risks?*

What students need to learn	Suggested teaching and learning
■ The complex causes of health risks.	■ Investigating the different causes of health risk, including pollution.
■ The relationship between socio-economic status and health.	■ Researching how closely linked health patterns are with spatial variations in economic standards of living.

■ The links between some diseases and geographical features.	■ Developing an awareness of how factors such as geographical distance and socio-economic status may greatly influence access to health care globally, nationally and/or at small scales. ■ Exploring how the development and spread of infection are often linked to geographical features (eg transport routes, accessibility, environment, water quality) and may follow geographical pathways and patterns.
■ Models that may help in the understanding of health risk causes and patterns.	■ Models, such as diffusion models, may help in understanding these causes and patterns.

3 Pollution and health risk

Enquiry question: *What is the link between health risk and pollution?*

What students need to learn	Suggested teaching and learning
■ The link between different pollution types and the health of societies.	■ Researching the differing types and sources of pollution and the direct threats and effects on health, as well as indirect effects to any society economically.
■ The relative health risks associated with incidental and sustained pollution.	■ Weighing up the difference to any society's health between one-off incidents such as Bhopal, Chernobyl and Harbin; and longer-term sustained pollution such as ozone depletion and skin cancer, and global warming and malaria spread.

■ The link between pollution, economic development and changing health risks.	■ Examining how types and amounts of pollution change with economic development and how, for example, the Kuznet curve and epidemiological models can assist in explaining these factors.
■ The role of pollution fatigue to reducing health risk.	■ Investigating the link between pollution fatigue and the backlash from the public to pressurise for effective management.

4 Managing the health risk

Enquiry question: *How can the impacts of health risk be managed?*

What students need to learn	Suggested teaching and learning
■ The socio-economic and environmental impacts of health risk.	■ Investigating the varying impact of infection and disease on people and communities and the resultant short- and long-term economic burdens on society.
■ How health risk impacts have led to differing management strategies and policies.	■ Developing an awareness of why some health risks and environments are harder to manage than others; such as indirect spin-offs from modern societies and consumerism like depression, obesity and diabetes.
■ The different agencies involved in health risk, especially international efforts.	■ Examining the increasing role of international agencies (official TNCs and NGOs) in managing health risk and pollution.

■ Which health risks can be managed effectively and which cannot; and the role of sustainability.	■ Researching the needs for longer-term sustainable planning for a better health environment; while shorter-term planning may also be necessary in some circumstances. Contrasting scenarios such as AIDs and emergency disease relief could be used.
---	---

Suggested fieldwork opportunities for pollution and human health at risk	
Local fieldwork ■ Using GIS and primary surveys to look at patterns of mortality, immortality and disease and the causal factors at a local scale and inequality of access to health care. ■ Surveys of selected diseases and problems, eg drug addiction, obesity, cancer, to study patterns of occurrence ie introduction to epidemiology. ■ Surveys of types of pollution eg water, air etc using primary and secondary data to identify areas of risk.	Residential and long-haul fieldwork ■ LEDC fieldwork could include surveys of health issues and health care access eg in Morocco, Kenya etc. ■ Fieldwork in distant locations to assess levels of pollution (land, air, water, noise, visual etc) for example in large urban areas or within a river catchment, or coastally, eg in Poland (acid rain impact survey).